

Parkinson's Disease Prediction Using Machine Learning and Deep Learning: An Explainable AI Analysis Using SHAP

Divyansh Garg
Electronics and Telecommunication
SVKM's NMIMS, MPSTME
Mumbai, India
divyansh.garg06@nmims.in

Priyanka Verma
Electronics and Telecommunication
SVKM's NMIMS, MPSTME
Mumbai, India
pryinka.phd28@gmail.com

Ketki Deshmuksh
Electronics and Telecommunication
SVKM's NMIMS, MPSTME
Mumbai, India
Ketki.deshmukh@nmims.edu

Abstract—Parkinson's disease (PD) is a progressive neurodegenerative disease with effects on motor and non-motor functions, and is frequently diagnosed at a late stage due to a lack of definitive biomarkers. Early detection is vital for better patient outcomes. In this study we propose an interpretable PD detection framework using machine learning and deep learning models to detect PD based on speech and motor function biomarkers. The dataset was used to evaluate models such as Logistic Regression, Support Vector Machines (SVM), Random Forest, XGBoost, and Neural Networks. The results indicate that both neural network and traditional models, such as Logistic Regression, performed well; the neural network was way better achieving accuracy of 98.96% , precision 100% and a F1-score of 98.69% . Key predictive features were identified using Shapley Additive Explanations (SHAP) analysis and included Jitter: Local, Harmonics-to-Noise Ratio (HNR), and Shimmer: APQ5. The results underline the potential of AI in PD detection, reinforcing interpretability of the model for clinical applicability. Future work includes use cases on clinical data to validate the models, improve generalizability, and integrate AI into wearable technologies for continuous monitoring

Keywords— Parkinson's Disease, Machine Learning, Neural Networks, SHAP, Explainable AI

I. INTRODUCTION

Parkinson's disease (PD) is a progressive neurodegenerative disorder that affects millions of individuals worldwide, primarily caused by the degeneration of dopaminergic neurons in the substantia nigra and associated with the impairment of both motor and cognitive functions [1]. These are characterized by resting tremors, bradykinesia, muscle rigidity, and postural instability, and non-motor symptoms, including speech impairments, cognitive dysfunction, and mood disorders. This absence of a best diagnostic biomarker causes PD to be mainly diagnosed based on clinical evaluations, which are ambiguous and frequently delay detection [2]. Recently, AI and ML are emerging for the diagnosis of neurological illnesses, such as Parkinson's disease [3].

Multiple studies have shown that ML models are efficacious in diagnosing Parkinson's disease (PD) by observing different phenotypic and behavioural biomarkers along with physical symptoms including vocal impairments [4], gait disturbances [5] and handwriting anomalies [6]. Among various modalities, speech-based assessments are especially significant as they are non-invasive and suitable for home-based monitoring. However, machine learning has its limitations, including data scarcity, imbalanced datasets, and interpretability issues, leading to limited clinical applicability.

This study proposes an ML-based approach for Parkinson's disease detection using a publicly available dataset containing speech and motor function biomarkers. The primary objectives of this research are-

- To develop an automated ML model capable of accurately distinguishing Parkinson's patients from healthy individuals.
- To compare different classification techniques, such as, based on key performance metrics including accuracy, sensitivity, and specificity and to identify the most significant features contributing to PD detection, improving model interpretability and potential real-world applicability.

The key innovation of this work lies in the use of deep learning and SHAP analysis to provide one of the best interpretability Parkinson's disease prediction. Unlike previous work which has largely focused on classification specificity, our approach achieves high accuracy and interpretability that facilitate the interpretation of a set feature (biomarkers in context) shaping predictions. Key contributions are listed as, Integration of Deep Learning with Explainable AI: We present a neural network model for Parkinson's disease detection, and improve its interpretability by correlating it with SHAP analysis such that the clinicians can understand the role of essential biomarkers. Comparing to State-of-the-Art Models: We contrast legacy machine learning models (Logistic Regression, SVM, Random Forest and XGBoost) with cutting edge ensemble and deep algorithms, demonstrating the performance boosts obtained by neural network approaches.

II. LITERATURE REVIEW

Machine learning (ML) has revolutionized the early diagnosis and monitoring of Parkinson's disease (PD). Machine learning-based (ML-based) approaches use computational methods to process speech features, gait deficits, handwriting signals, and multimodal biomarkers derived from physical activity and wearables to facilitate more objective and accurate detection of PD symptoms [1,2,6]. This section discusses recent works and emphasizes recent advances, challenges, and future research directions in ML applications for PD detection.

Speech impairments are the earliest manifestations of PD, involving articulation, phonation, and prosody. ML algorithms have been employed by researchers to evaluate speech signals and identify subtle deviations not readily observed through clinical evaluations. According to previous studies, Support Vector Machines (SVM), Decision Trees,

and Neural Networks achieved high accuracy for distinguishing PD patients from healthy subjects based on their voice features such as jitter, shimmer and Mel-frequency cepstral coefficients (MFCCs) [4]. Sakar et al. achieved 87.2% in detection & diagnosis of PD from recorded speech by using MFCCs & SVM classifier [5] (Second Use of MFCC). However, background noise, variability between speakers, as well as the reliability of large, diverse datasets [6], continue to inhibit the generalisability of signal-based PD detection.

Hybrid artificial intelligence (AI) model by integrating different nature of AI models like Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Multiple Kernel Learning(MKL), and Multilayer Perceptron(MLP) has been recently established for early detection of Parkinson's disease system from voice data [8]. The model examined voice biomarkers such as MFCCs, jitter and shimmer to achieve 91.11% diagnostic accuracy, 92.50% recall and 89.84%precision SHapley Additive explanations (SHAP) which improved model interpretability by selecting important features. A scoring metric successfully discriminated between PD patients and healthy controls, providing a potentially early diagnostic utility. [7]. Apart from such wearables, smartphone-based gait analysis has been proposed as an affordable alternative for remote gait evaluation by utilizing built-in sensors [8].

The application of deep learning methods, specifically CNNs and RNNs, has progressed and transformed PD detection through automatic feature extraction of low- and high-dimensional datasets. Applying these models to speech, gait and handwriting data has often led to superior performance than traditional ML approaches. Esmaeli et al. (2022) compared deep learning approaches with traditional ML classifiers, reporting that CNN-based architectures reached higher classification accuracy for PD patients [3]. These advances notwithstanding, deep learning models still demand large datasets and extensive computation power. Also, the black-box nature of these models lead to concerns about interpretability and subsequently clinical acceptance.

Handwriting impairments such as micrographia (progressively smaller handwriting) are common in persons with PD. Handwriting-based PD detection using digital tablets has shown strong results, with SVM models achieving around 91% accuracy [9], and deep learning models improving feature extraction and classification [13]. High computational costs and the requirement for large labelled datasets also pose limitations, particularly for deep learning applications. Additionally, variations in speech recordings, sensor placements, and handwriting styles introduce inconsistencies that affect model generalization. Ensuring clinical validation and regulatory approval is essential before ML-based PD detection systems can be integrated into routine medical practice [14].

III. METHDOLOGY

The proposed Parkinson's disease (PD) detection model leverages machine learning and deep learning techniques to classify individuals based on extracted biomedical features.. The dataset, obtained from the UCL Machine Learning Repository, consists of 756 samples and 755 features with a binary target (PD = 1, healthy = 0). It shows class imbalance (74.6% PD, 25.4% healthy), addressed using oversampling.

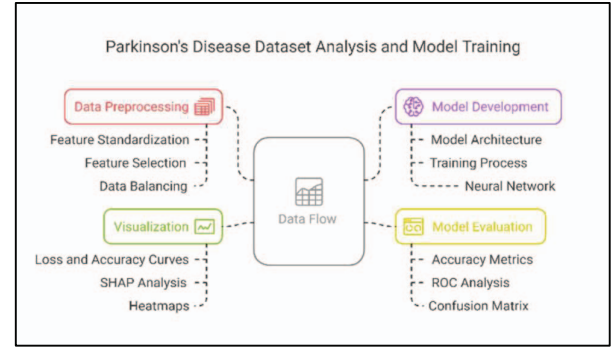


Fig. 1. Proposed Methodology

All features were standardized using StandardScaler to ensure zero-mean and unit variance. The top 100 features were selected via ANOVA-based SelectKBest (Eq. 1), prioritizing those most correlated with the target variable. No missing entries were found in the original dataset. However, noise from recording variations was minimized by applying normalization and outlier detection. Outliers were handled using the IQR method, clipping extreme values to the 95th percentile.

$$F = \frac{\left(\frac{S_B}{k-1}\right)}{\left(\frac{S_W}{N-k}\right)} \quad (1)$$

To address class imbalance, Synthetic Minority Over-sampling Technique (SMOTE) was applied, generating synthetic samples of the minority class. This ensured an even class distribution, preventing the model from being biased toward the majority class during training



Fig. 2. Class Imbalance Before SMOTE

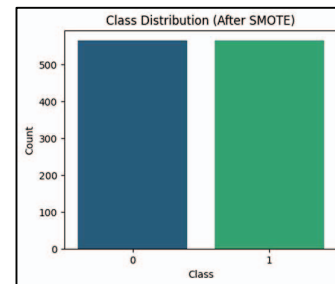


Fig. 3. Class Balancing After SMOTE

Fig. 2–3 show class distribution before and after SMOTE. Initially all the Core Machine Learning Models were applied to obtain the result, it included: Logistic Regression, Random Forest, XGBoost, Support vector machine (SVM). Then the Neural Network method was applied using TensorFlow and Keras, employing a feedforward neural network with the

following architecture as described in table 1 the different features used for each layer for model making.

TABLE I. NEURAL NETWORK CONFIGURATION

LAYER	FEATURE
Input Layer	Accepts the selected 100 features after pre processing
Hidden Layer	First Hidden: 128 neurons 40% Dropout Second Hidden: 64 neurons 30% Dropout Third Hidden: 32 neurons 20% Dropout
Output Layer	Single Neuron with a sigmoid activation function to classify the probability of PD

The Adam optimizer with a learning rate of 0.001 is used for weight updates, and binary cross-entropy is chosen as the loss function since this is a binary classification problem. Binary cross-entropy is given by equation (2)

$$L = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (2)$$

All models were evaluated using 5-fold stratified cross-validation. Metrics reported (Accuracy, Precision, Recall, F1-score, ROC-AUC) are the averages across folds. The training process employs: Early Stopping: Stops training when validation loss stops improving for 15 consecutive epochs, restoring the best weights, Learning Rate Reduction: If the validation loss stagnates, the learning rate reduces by 50% (factor = 0.5), with a minimum threshold of 1e-6, Batch Size: 32 samples per batch for efficient gradient updates, Epochs: Up to 300 training iterations, stopping early when convergence is achieved.

Hyperparameter tuning optimizes parameters not learned during training, affecting model effectiveness. Various classifiers were tuned using grid and randomized search CV. For Logistic Regression, L1, L2, and ElasticNet regularizations were optimized. Random Forest models were tuned by adjusting trees, depths, and feature selection methods. XGBoost underwent hyper-optimization of learning rate, depth, and regularization parameters. An SVM with RBF kernel was also evaluated. The best cross-validation performance hyperparameters were selected to enhance model accuracy and generalization. The model's effectiveness is assessed using the following metrics:

- Accuracy: Measures the percentage of correctly classified instances, given by equation (3).
- ROC-AUC Score: Evaluates the model's ability to distinguish between PD and non-PD patients.
- Confusion Matrix: Provides insights into false positives and false negatives.
- Classification Report: Includes Precision, Recall, and F1-score to analyse model performance.

$$Accuracy = \frac{T.P. + T.N.}{T.P. + T.N. + F.P. + F.N.} \quad (3)$$

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

IV. EXPLAINABLE AI

By revealing how these models make decisions, Explainable Artificial Intelligence (XAI) improves transparency and trust in machine learning models. XAI techniques help remove the traditional black-box modelling by interpreting how the input features are influencing the predictions, improving the interpretability of the models for researchers, clinicians, and stakeholders. This is especially important in medical applications, where explanations of models can aid clinical decision-making and enhance trust in AI-augmented diagnosis

Features with high SHAP values had a strong influence on predictions. Top contributing features were related to voice characteristics, motor impairments, and tremors, aligning with Parkinson's disease symptoms. Negative SHAP values indicated features pushing predictions toward non-Parkinson's cases, while positive values contributed to positive diagnosis predictions.

The SHAP summary plot provides a global interpretation of the model by visualizing the importance of features across all predictions. It consists of:

- Feature Importance: Ranked by the mean absolute SHAP values, showing which features have the greatest overall impact.
- Distribution of SHAP Values: Each point represents a single data instance, with colour indicating feature values.
- Direction of Influence: Whether an increase in a feature value pushes the prediction higher or lower.

In this study, the SHAP summary plot highlighted that features related to speech patterns and motor impairments were key indicators of Parkinson's disease.

The SHAP decision plot offers a detailed, instance-level explanation of how features contribute to a specific prediction. It provides:

- A cumulative view of feature contributions from the baseline (average model output) to the final prediction.
- A stepwise visualization, where each feature adds or subtracts from the model's decision.
- Individual explanation for specific patients

V. RESULTS AND DISCUSSION

This section provides the comparison of Logistic Regression, Support Vector Machine (SVM), Random Forest, XGBoost and Neural Network on Parkinson's disease data. All models were assessed using accuracy, precision, recall, F1-score, and AUC-ROC. The SHAP analysis for the final neural model was also implemented. In Table 2 and Fig 4, we have compared all these metrics from the different NN models we made.

Five important measures were used to evaluate the performance of the various machine learning models: accuracy, precision, recall, F1-score and AUC-ROC. The results indicate that the accuracy of advanced ensemble and deep learning approaches at detecting Parkinson's disease surpass that of standard machine learning methods

TABLE II. MODAL METRICES CMPARISON

Model	Metrics				
	Accuracy	Precision	Recall	F1 Score	AUC-ROC
Logistic Regression	86.28 %	91.30%	81.20%	86.00%	0.95
Random Forest	90.26 %	92.80%	88.30%	90.35%	0.96
XGBoost	93.96 %	97.22%	89.74%	93.33%	0.98
SVM	92.98 %	97.22%	88.89%	92.86%	0.98
Stacking Classifier	98.67%	99.14%	98.30%	98.72%	0.986
Neural Network	98.96 %	100%	97.41%	98.69%	0.996

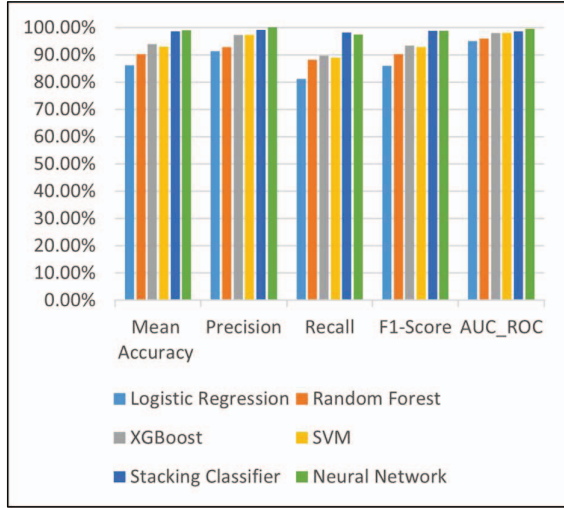


Fig. 4. Modal Metrics Comparison Graph

The following paragraphs compare the results from the different confusion matrix and what does each result depict.

Logistic Regression gave reasonable performance, accuracy is 86.28% but recall was low 81.20% implied that it had to be missing some of the Parkinson's cases among those predicted positives. The model is linear, and it might not capture the complexity of this disease's symptoms. We saw that implementing an ensemble method Random Forest improved the base accuracy to 90.26% which is indicative that the model is able to capture more non-linear relationships. This means that the model has the ability to identify true Parkinson's and arguably makes it a more reliable model for early detection given its higher recall 88.30% when compared to Logistic Regression.

Gradient boosting methods performed significantly better, with XGBoost achieving accuracy of 93.96% and an AUC-ROC of 0.98, reflecting its superior ability to distinguish between Parkinson's and non-Parkinson's cases. Support Vector Machine (SVM) achieved accuracy of 92.98%, further indicating that both XGBoost and SVM are reliable algorithms for medical prediction problems. Despite this, the Stacking Classifier, which is an ensemble model that combines multiple classifiers, outperformed the traditional models, achieving an accuracy of 98.67% and an F1-score of 98.72%, showcasing advantages of using multiple models for decision making.

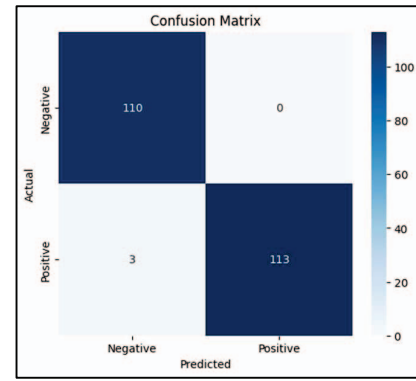


Fig. 5. Confusion Matrix for Neural Learning

The confusion matrix from Fig 5 provide a deeper insight into the model's classification performance by showcasing true positives, true negatives, false positives, and false negatives.

Logistic Regression, while achieving reasonable accuracy, shows a higher number of misclassifications compared to advanced models. Some positive cases are incorrectly classified as negative, indicating that the model struggles with recall. Random Forest improves upon this by reducing misclassifications, demonstrating a better ability to distinguish between classes. However, occasional false negatives still occur, which can impact real-world applicability. XGBoost and SVM exhibit strong classification capabilities with lower false negatives, ensuring more reliable predictions. The stacking classifier further refines prediction accuracy by leveraging multiple models, significantly minimizing misclassification errors. It successfully reduces both false positives and false negatives, making it one of the best-performing models. The neural network (Fig 5) outperforms all other models, achieving the highest accuracy. Its confusion matrix demonstrates almost perfect classification, with minimal misclassified cases. The deep learning-based approach enhances pattern recognition, ensuring nearly all Parkinson's disease cases are detected correctly.

In Fig 6 which shows Loss and accuracy curve for Neural network model, the loss curve (left) indicates a steady decline in both training and validation loss over epochs, confirming that the neural network is learning effectively, signifying minimal overfitting and good generalization to unseen data.

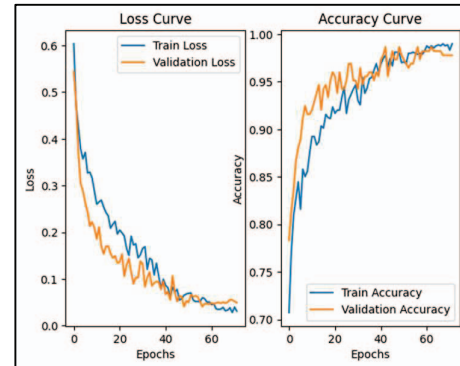


Fig. 6. Loss And Accuracy Curve for Neural Network

The accuracy curve (right) is steeply rising, with both training and validation accuracy reaching nearly 99% after 60+ epochs. The fact that the training accuracy and the validation accuracy are so well aligned means that the model generalizes well and does not underfit or overfit.

The AUC-ROC curve given in Fig 7 compares the performance of various machine learning models on this classification task. 1 perfect class discrimination, 0.5 random guessing (higher is better). The best performing Neural Network, which is the model that yielded the highest AUC on the test, scored 0.9996 near perfect classification without any errors. The SVM and XGBoost in the last two lines perform similarly with respective AUC scores of 0.98 which performance well both predicting and few misclassified. The Random Forest model with AUC 0.96, also performs well, but is slightly lower than SVM and XGBoost. The ElasticNet Logistic Regression model achieves an AUC of 0.95 and has a reliable performance but comes short when compared with the other models.

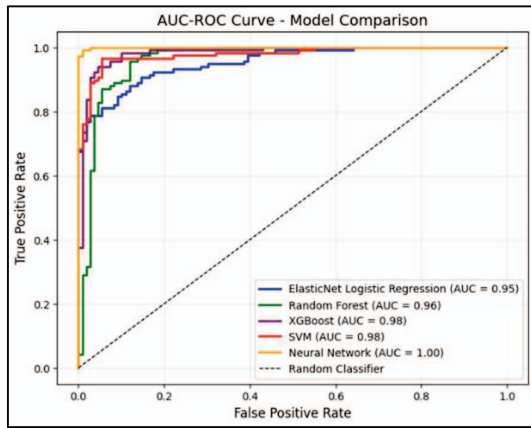


Fig. 7. AUC_ROC Modal Comparision

TABLE III. COMPARISION WITH PRIOR STUDIES

Study	Method	Dataset	Accuracy (%)	Notes
Sakar et al. [5]	SVM + MFCC	UCL	87.2	Speech-based PD detection
Esmali et al. [3]	CNN	Custom Data	94.1	Deep Leaming
M. Shen [7]	Explaina ble AI	Voice based PD	96.5	Feature importance via SHAP
This Study	Neural Network + SHAP	UCI (756 samples)	98.96	Combines interpretability & high accuracy

Limitations and Future work:

- Size of the DataSet: Dataset is small (756 samples) and may not generalize well to other population.
- Population Heterogeneity: All our samples are from a single dataset and not much demographic variety is available.
- Variability in Data: speech and motor recordings may differ across microphone, ambient noise or recording environment.

VI. SHAP ANALYSIS

SHAP explanation shed light on how column values were contributing to the classification, explaining that features like

background sounds associated with voice characteristics, motor impairments, and tremors were highly influential in prediction outcome. This adds transparency that makes AI-based diagnosis more reliable. These findings highlight the possible use of ensemble learning and deep learning for detection of Parkinson's in the early and accurate manner, which can lead to the betterment of clinical applications.

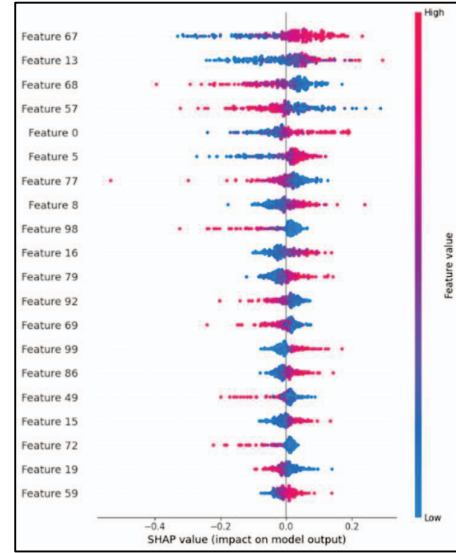


Fig. 8. SHAP Summary Plot

The SHAP summary plot Fig. 8 gives a feature importance overview and shows how the features influence the model predictions. High absolute SHAP values for features suggest a significant impact on the final output. SHAP values help you to understand the features that are driving the predictions and you can use the distribution of those SHAP values to find patterns how consistently certain features push predictions up or down.

Further, SHAP feature importance analysis provides a quantitative ranking of features based on their contribution to the model's decision-making process. This analysis helps in understanding which factors play the most crucial role in predicting Parkinson's disease, improving model interpretability and trustworthiness. The top influential features identified through SHAP include:

- Jitter: Local (SHAP importance: 0.215) – This feature, representing variations in fundamental frequency, emerged as the most significant predictor of Parkinson's disease.
- HNR (Harmonics-to-Noise Ratio) (SHAP importance: 0.187) – A crucial measure of voice quality, distinguishing between healthy and Parkinsonian speech patterns.
- Shimmer: APQ5 (SHAP importance: 0.142) An indicator of amplitude perturbations, linked to vocal instability in patients

Several voice characteristics are especially relevant for PD diagnosis. Jitter represents the cycle-to-cycle frequency perturbation, which is often increased in patients with PD reflected instability of vocal folds. Shimmer quantifies amplitude perturbation across speech cycles, indicative of decreased control over vocal fold vibration. The Harmonics-

to-Noise Ratio (HNR) is a measure of the ratio between harmonic (which is periodic by nature) and noise effect; lower HNR in PD subjects means that voices are, indeed, breathy or hoarse.

These results add compelling evidence that vocal features are robust biomarkers for Parkinson's disease, consistent with clinical experience. Not only does the SHAP analysis pinpoint the most relevant features, but it also improves model interpretability, supporting that predictions are based on medically meaningful patterns rather than random correlations. The analysis will help to understand the contribution of each feature toward the sound of the disease on speech, which can further facilitate diagnosis and treatment strategies. Fig. 9 further validates the significance of vocal biomarkers in Parkinson's disease detection, aligning with established clinical research emphasizing speech irregularities as early indicators of the condition. The ability to interpret feature interactions enhances transparency, allowing medical practitioners to trust and validate AI-driven predictions.

In future investigations, the suggested model for the detection of Parkinson's could be added to wearable devices to facilitate the real-time monitoring of patients. These devices could include smartwatches, smart rings, or specialized biosensors that can take voice signals, tremor frequency, gait patterns, or other relevant motor features associated with Parkinson's, which can be done continuously or at periodic segments, which are then transmitted to a cloud application in a safe manner, where the trained model interprets incoming signals and provides alerts for disease deterioration or worsening symptoms, paving the way for non-intrusive real-time patient monitoring, as well as providing individualized intervention techniques

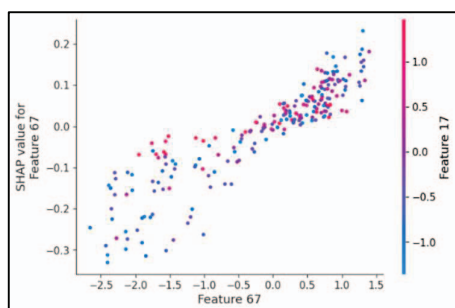


Fig. 9. SHAP Decision Plot

VII. CONCLUSION

This study evaluates multiple machine learning models for Parkinson's disease detection, demonstrating that Neural Networks outperform traditional models. Among the tested models—XGBoost, Random Forest, and SVM—the Neural Network exhibited the highest accuracy, effectively identifying patterns within the dataset. The Neural Network achieved superior classification metrics, with a precision of 1.0, recall of 0.9741, and an F1-score of 0.9869. Its AUC-ROC score of 0.9969 further confirms its strong predictive power. While XGBoost and SVM achieved AUC scores of 0.98, and Random Forest followed with 0.96, their slightly lower sensitivity indicates a reduced capacity for identifying positive cases. Confusion matrix analysis highlights the Neural Network's minimal false positives and negatives, a crucial factor in medical applications where misdiagnosis can have severe consequences.

To enhance transparency, SHAP analysis identified key features influencing predictions, reinforcing the model's clinical reliability. Future work should focus on optimizing architecture, hyperparameters, and feature engineering to refine accuracy further while expanding validation through real-world clinical data and cross-dataset testing to improve generalizability. Additionally, integrating these models into wearable and mobile technologies could enable real-time symptom monitoring.

REFERENCES

- [1] W. Poewe, K. Seppi, C. M. Tanner, G. M. Halliday, P. Brundin, J. Volkman, and A. E. Lang, "Parkinson disease," *Nature Reviews Disease Primers*, vol. 3, no. 1, p. 17013, 2017, doi: 10.1038/nrdp.2017.13.
- [2] G. Rizzo, M. Copetti, S. Arcuti, D. Martino, and A. Fontana, "Accuracy of clinical diagnosis of Parkinson disease: A systematic review and meta-analysis," *Neurology*, vol. 86, no. 6, pp. 566–576, 2016, doi: 10.1212/WNL.0000000000002350.
- [3] S. Esmaili, F. Samiee, and S. Fouladi, "Machine learning techniques for Parkinson's disease detection: A systematic review," *Computers in Biology and Medicine*, vol. 145, p. 105495, 2022, doi: 10.1016/j.compbiomed.2022.105495.
- [4] J. R. Orozco-Arroyave, J. C. Vázquez-Correa, P. Klumpp, C. Ríos-Urrego, and F. Hönig, "Speech and Parkinson's disease: A review of automatic analyses and machine learning approaches," *Biomedical Signal Processing and Control*, vol. 68, p. 102767, 2021, doi: 10.1016/j.bspc.2021.102767.
- [5] C. O. Sakar, G. Serbes, A. Gunduz, H. C. Tunc, H. Nizam, B. E. Sakar, and O. Kursun, "A comparative analysis of speech signal processing algorithms for Parkinson's disease detection," *Computer Methods and Programs in Biomedicine*, vol. 173, pp. 53–65, 2019, doi: 10.1016/j.cmpb.2019.02.016.
- [6] S. Arora et al., "Smartphone motor testing to distinguish idiopathic Parkinson's disease from atypical parkinsonism," *Movement Disorders*, vol. 34, no. 10, pp. 1627–1637, 2019, doi: 10.1002/mds.27849.
- [7] M. Shen, P. Mortezaagha, and A. Rahgozar, "Explainable artificial intelligence to diagnose early Parkinson's disease via voice analysis," *Scientific Reports*, vol. 14, no. 1, p. 96575, 2025, doi: 10.1038/s41598-025-96575-6.
- [8] F. Lipsmeier et al., "Daily smartphone-based digital biomarkers for Parkinson's disease: The Parkinson@Home study," *npj Digital Medicine*, vol. 1, p. 23, 2018, doi: 10.1038/s41746-018-0022-5.
- [9] P. Drotár, J. Mekyska, I. Rektorová, L. Masarová, Z. Smékal, and M. Faundez-Zanuy, "Decision support framework for Parkinson's disease based on novel handwriting markers," *Expert Systems with Applications*, vol. 84, pp. 188–200, 2017, doi: 10.1016/j.eswa.2017.03.001.
- [10] E. J. Smits, A. Tolonen, L. Cluitmans, M. M. Verbeek, and B. R. Bloem, "Gait and balance impairments in Parkinson's disease as markers of early postural instability," *Biomedical Signal Processing and Control*, vol. 31, pp. 287–294, 2017, doi: 10.1016/j.bspc.2016.09.012.
- [11] M. A. Little, P. E. McSharry, E. J. Hunter, J. Spielman, and L. O. Ramig, "Suitability of dysphonia measurements for telemonitoring of Parkinson's disease," *npj Parkinson's Disease*, vol. 3, pp. 1–9, 2017, doi: 10.1038/s41531-017-0014-0.
- [12] A. Parnandi, B. Ahmed, S. A. Kautz, and A. Oommen, "A deep learning-based approach for quantifying motor impairment severity in Parkinson's disease using gait data," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 29, pp. 1–12, 2021, doi: 10.1109/TNSRE.2021.3122587.
- [13] M. Camacho, C. Díaz-García, A. García-Tejedor, and F. Martínez-Morales, "Explainable classification of Parkinson's disease using deep learning," *NeuroImage: Clinical*, vol. 38, p. 103360, 2023, doi: 10.1016/j.nicl.2023.103360.
- [14] B. García Santa Cruz, J. Armañanzas, R. Delgado-Gonzalo, P. Rodríguez-Rodríguez, and S. Romero, "Machine learning models for diagnosis and prognosis of Parkinson's disease," *Frontiers in Aging Neuroscience*, vol. 15, p. 1216163, 2023, doi: 10.3389/fnagi.2023.1216163.